S6 — Advanced Organic Chemistry and Analysis

6.1 Aromatic Chemistry

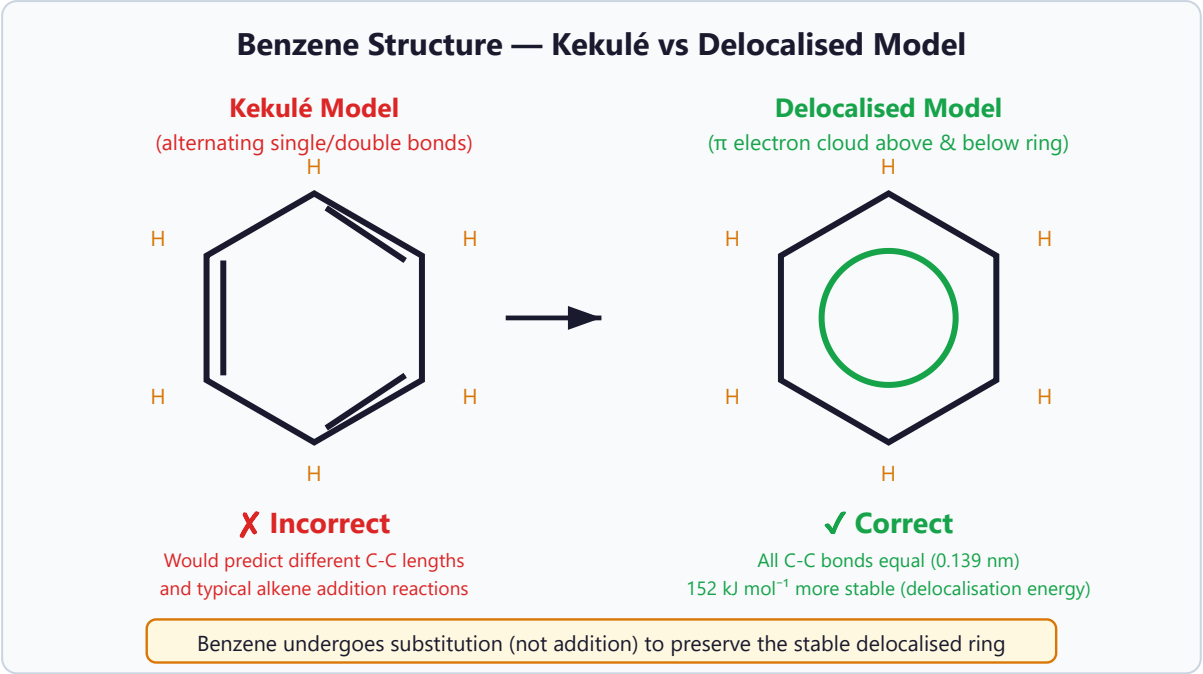
Structure of Benzene

Benzene (C_6H_6): A planar, cyclic hydrocarbon with a ring of six carbon atoms. Each carbon is sp^2 hybridised with bond angles of 120°. The unhybridised p orbital on each carbon overlaps sideways with those on adjacent carbons, forming a delocalised π system above and below the plane of the ring.

Kekulé model: Originally proposed as alternating single and double bonds (cyclohexatriene). However, this model is inconsistent with experimental evidence:

- All C−C bond lengths in benzene are equal (0.139 nm), intermediate between a single bond (0.154 nm) and a double bond (0.134 nm).
- Benzene does not undergo addition reactions typical of alkenes (e.g. it does not readily decolourise bromine water).
- The enthalpy of hydrogenation of benzene (−208 kJ mol^{-1}) is less exothermic than predicted for three C=C bonds (3 × −120 = −360 kJ mol^{-1}), showing benzene is more stable than the hypothetical cyclohexatriene by about 152 kJ mol^{-1} (the delocalisation/resonance energy).

Delocalised model: The six π electrons are spread evenly over all six carbon atoms in a ring of electron density. This delocalisation makes benzene more thermodynamically stable than the Kekulé structure and explains its tendency to undergo substitution rather than addition reactions (preserving the stable delocalised ring).



Electrophilic Substitution of Benzene

The delocalised π electron ring is a region of high electron density that attracts electrophiles. However, benzene undergoes substitution rather than addition to preserve the stability of the aromatic ring.

Nitration:

$$C_6H_6 + HNO_3 \longrightarrow (\text{conc. } H_2SO_4, 50°C)\ C_6H_5NO_2 + H_2O$$

The electrophile is the nitronium ion (NO_2^+), generated by the reaction of concentrated nitric acid with concentrated sulfuric acid: $HNO_3 + H_2SO_4 \rightarrow NO_2^+ + HSO_4^- + H_2O$. The mechanism involves the NO_2^+ being attracted to the π system, forming a positively charged intermediate (Wheland intermediate), followed by loss of H^+ to restore aromaticity. Temperature must be kept below 55 °C to prevent dinitration.

Halogenation:

$$C_6H_6 + Br_2 \longrightarrow (AlBr_3 \text{ or } FeBr_3 \text{ catalyst})\ C_6H_5Br + HBr$$

A halogen carrier catalyst (Lewis acid such as $AlBr_3$, $FeBr_3$, or $AlCl_3$) is needed to generate the electrophile. The catalyst polarises the Br–Br bond or forms Br^+: $Br_2 + AlBr_3 \rightarrow Br^+ + AlBr_4^-$. The Br^+ then attacks the π ring.

Friedel-Crafts alkylation:

$$C_6H_6 + RCl \longrightarrow (AlCl_3)\ C_6H_5R + HCl$$

The $AlCl_3$ catalyst generates a carbocation electrophile (R^+) from the halogenoalkane. This introduces an alkyl group onto the benzene ring.

Friedel-Crafts acylation:

$$C_6H_6 + RCOCl \longrightarrow (AlCl_3)\ C_6H_5COR + HCl$$

The electrophile is an acylium ion (RCO^+). This introduces an acyl group (C=O bonded to R) onto the ring. Acylation is preferred over alkylation in synthesis because it does not give multiple substitutions (the C=O group deactivates the ring).

Exam Tip: In all electrophilic substitution mechanisms on benzene, draw two curly arrows: one from the π ring to the electrophile (forming the new bond), and one from the C−H bond to the ring (restoring aromaticity by losing H^+). Show the intermediate Wheland complex with its positive charge delocalised over the ring.

6.2 Carbonyl Compounds

Aldehydes and Ketones

Both contain the C=O (carbonyl) group. Aldehydes have the carbonyl at the end of the chain (RCHO); ketones have it within the chain (RCOR'). The C=O bond is polar (δ+ on C, δ− on O) because oxygen is more electronegative than carbon, making the carbonyl carbon susceptible to nucleophilic attack.

Nucleophilic Addition

The key mechanism for carbonyl compounds. A nucleophile attacks the δ+ carbon of the C=O group:

With HCN (hydrogen cyanide): The nucleophile is CN^-. It attacks the δ+ carbon, breaking the π bond and forming an alkoxide intermediate, which is then protonated by HCN to give a hydroxynitrile (cyanohydrin). This reaction extends the carbon chain by one carbon atom and introduces both an OH and CN group. A trace of base (NaCN or KCN) is used as a catalyst to generate CN^-.

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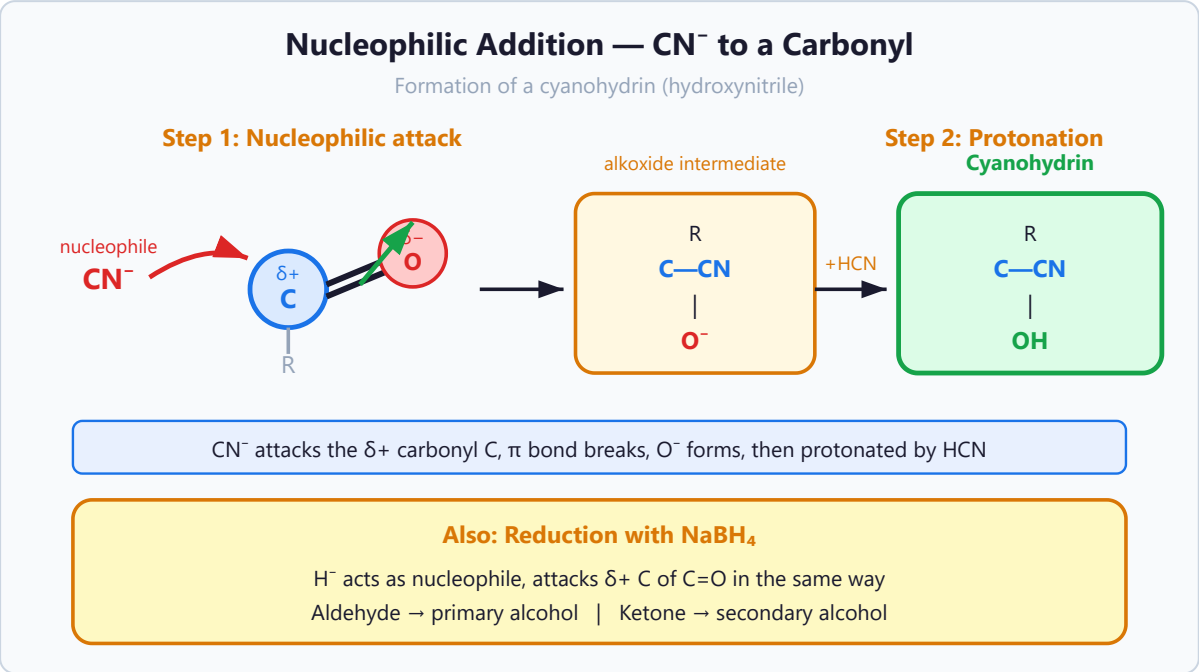
With $NaBH_4$ (sodium borohydride): A reducing agent. The H^- ion acts as the nucleophile, attacking the δ+ carbonyl carbon. Aldehydes are reduced to primary alcohols; ketones are reduced to secondary alcohols. $NaBH_4$ is used in aqueous/ethanolic solution.



Distinguishing Aldehydes from Ketones

Test	Reagent	Aldehyde result	Ketone result
Tollens' test	$[Ag(NH_3)_2]^+$ (silver mirror reagent), warm	Silver mirror forms (Ag^+ reduced to Ag)	No reaction
Fehling's test	Fehling's solution (Cu^{2+} tartrate complex), heat	Brick-red precipitate of Cu_2O	No reaction
2,4-DNP test	2,4-dinitrophenylhydrazine (Brady's reagent)	Orange/yellow precipitate (confirms C=O present)	Orange/yellow precipitate (confirms C=O present)

Tollens' and Fehling's tests work because aldehydes can be oxidised to carboxylic acids, but ketones cannot be oxidised under these mild conditions. The 2,4-DNP test confirms the presence of a carbonyl group (aldehyde or ketone) but does not distinguish between them. The melting point of the 2,4-DNP derivative can be used to identify the specific aldehyde or ketone.

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6.3 Carboxylic Acids and Derivatives

Carboxylic Acids

Carboxylic acids contain the −COOH functional group. They are weak acids that partially dissociate in water: $RCOOH \rightleftharpoons RCOO^- + H^+$. They react with bases, carbonates, and reactive metals to form salts (carboxylates).

Esters

Esters (RCOOR') are formed by the condensation reaction between a carboxylic acid and an alcohol (Fischer esterification) in the presence of an acid catalyst (concentrated H_2SO_4). This is a reversible reaction.

Hydrolysis of esters:

- **Acid hydrolysis:** Refluxing with dilute acid (e.g. dilute H_2SO_4) breaks the ester back into the carboxylic acid and alcohol. This is reversible.
- **Base hydrolysis (saponification):** Refluxing with dilute NaOH produces the sodium salt of the carboxylic acid (carboxylate) and the alcohol. This is irreversible because the carboxylate ion is very stable.

Acyl Chlorides

Acyl chloride (RCOCl): A reactive derivative of carboxylic acids in which the OH of the COOH group is replaced by Cl. They are much more reactive than carboxylic acids because the C−Cl bond is weaker and the Cl is a good leaving group.

Acyl chlorides react vigorously at room temperature with:

Reagent	Product	Other product
Water (H_2O)	Carboxylic acid	HCl
Alcohol (ROH)	Ester	HCl
Ammonia (NH_3)	Amide ($RCONH_2$)	HCl

Amine ($R'NH_2$)	N-substituted amide (RCONHR')	HCl

These are all nucleophilic addition-elimination reactions. The nucleophile (H_2O, ROH, NH_3, or $R'NH_2$) attacks the δ+ carbonyl carbon, and the Cl^- leaves.

6.4 Amines and Amino Acids

Amines

Amine: An organic compound derived from ammonia (NH_3) by replacing one or more hydrogen atoms with alkyl or aryl groups. Primary amines have the formula RNH_2, secondary amines R_2NH, and tertiary amines R_3N.

Preparation of amines:

- Reaction of a halogenoalkane with excess ammonia in a sealed tube at elevated temperature: $RBr + 2NH_3 \rightarrow RNH_2 + NH_4Br$. Excess ammonia is used to minimise further substitution.
- Reduction of a nitrile with $LiAlH_4$ in dry ether or by catalytic hydrogenation: $RCN + 4[H] \rightarrow RCH_2NH_2$.
- Reduction of a nitro compound (important for aromatic amines): $C_6H_5NO_2 + 6[H] \longrightarrow$ (Sn/conc. HCl, reflux) $C_6H_5NH_2 + 2H_2O$.

Base properties of amines: Amines act as Brønsted-Lowry bases because the lone pair on nitrogen can accept a proton: $RNH_2 + H^+ \rightarrow RNH_3^+$. Aliphatic amines are stronger bases than ammonia because the electron-donating alkyl groups increase electron density on the nitrogen, making the lone pair more available for protonation. Aromatic amines (e.g. phenylamine, $C_6H_5NH_2$) are weaker bases than ammonia because the nitrogen lone pair is delocalised into the benzene ring, reducing its availability.

Amino Acids

Amino acid: An organic molecule containing both an amine group ($-NH_2$) and a carboxylic acid group (−COOH). The α-amino acids found in proteins have both groups attached to the same carbon atom (the α-carbon), along with a hydrogen atom and a variable side chain (R group).

Zwitterions: In solution at the isoelectric point, amino acids exist predominantly as zwitterions — the amine group is protonated ($-NH_3^+$) and the carboxylic acid group is deprotonated ($-COO^-$). This gives the molecule both a positive and a negative charge but an overall neutral charge. Zwitterions have high melting points and are soluble in water due to their ionic character.

In acidic solution, the COO^- group is protonated, giving a positive ion. In basic solution, the NH_3^+ group is deprotonated, giving a negative ion.

Peptide Bonds and Proteins

$$H_2NCHRCOOH + H_2NCHR'COOH \rightarrow H_2NCHRCONHCHR'COOH + H_2O$$

A peptide bond (−CONH−) forms by a condensation reaction between the −COOH of one amino acid and the $-NH_2$ of another, releasing water. Proteins are polypeptides formed from many amino acids linked by peptide bonds. Peptide bonds can be broken by hydrolysis (acid or base conditions, or enzymatically).

6.5 Polymers

Addition Polymers

Formed from alkene monomers by breaking C=C double bonds. No by-product is formed. Examples: poly(ethene), poly(propene), PVC, PTFE. They are non-biodegradable and chemically inert (strong C−C backbone).

Condensation Polymers

Condensation polymerisation: Monomers join together with the loss of a small molecule (usually water) at each linkage. The monomers must have two functional groups each (they are bifunctional).

Polyesters: Formed from a diol (HO−R−OH) and a dicarboxylic acid (HOOC−R'−COOH), or from a hydroxycarboxylic acid (HO−R−COOH). The linkage is an ester bond (−COO−). Example: Terylene (PET) from ethane-1,2-diol and benzene-1,4-dicarboxylic acid.

Polyamides: Formed from a diamine ($H_2N-R-NH_2$) and a dicarboxylic acid (HOOC−R'−COOH), or from an amino acid. The linkage is a peptide/amide bond (−CONH−). Examples: nylon-6,6 from 1,6-diaminohexane

and hexanedioic acid; Kevlar from 1,4-diaminobenzene and benzene-1,4-dicarboxylic acid.

Biodegradability: Condensation polymers are generally biodegradable because the ester or amide bonds can be hydrolysed by water, acids, bases, or enzymes. Addition polymers are generally non-biodegradable. Biodegradable polymers reduce landfill waste but may produce greenhouse gases (CO_2, CH_4) during decomposition.

Exam Tip: When drawing condensation polymer repeat units, remember to show the linkage (ester or amide bond) and indicate where water was lost. The repeat unit should be enclosed in brackets with continuation bonds on each side. Be able to identify the monomers from a given polymer structure by breaking the ester or amide bonds and adding back H and OH.

6.6 Organic Synthesis and Retrosynthesis

Organic synthesis: Planning a series of reactions to convert a starting material into a desired target molecule.

Retrosynthesis: Working backwards from the target molecule to identify simpler starting materials and the reactions needed. The target is disconnected at key bonds to identify precursor molecules.

Key synthetic routes to know (building a "reaction map"):

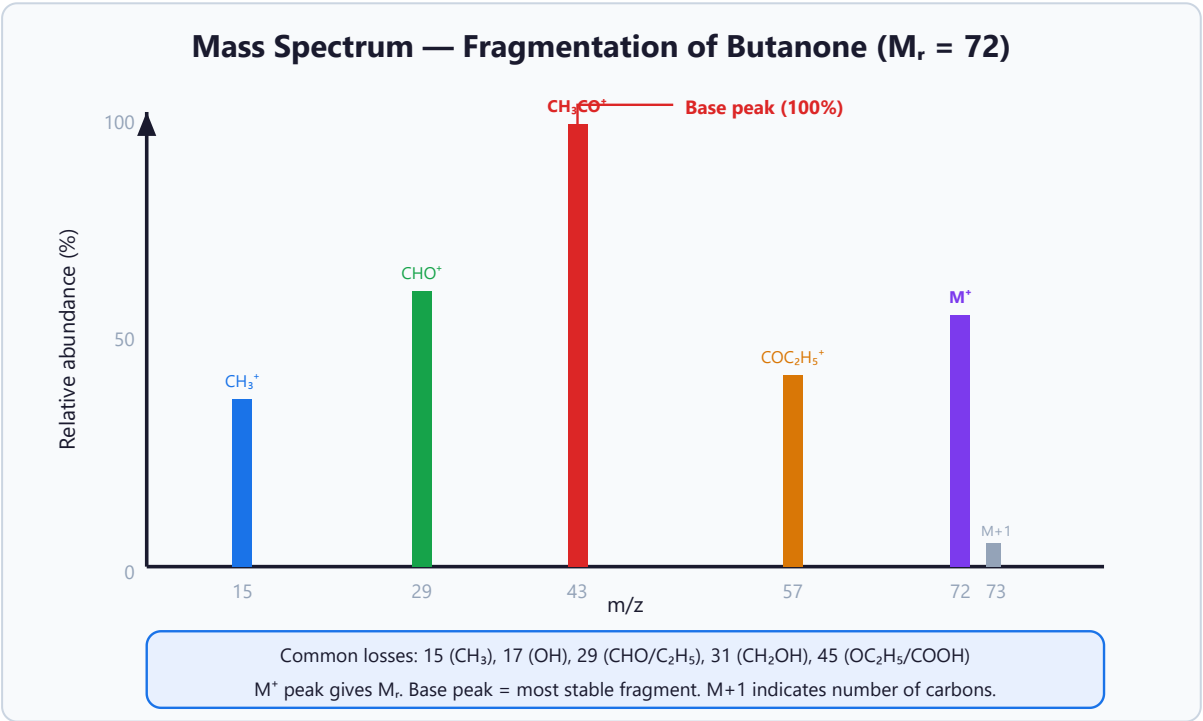
- Alkene → alcohol (hydration with H_3PO_4 catalyst, or addition of H_2O via H_2SO_4)
- Alcohol → halogenoalkane (HBr, or NaBr + H_2SO_4, or PCl_3/PCl_5/$SOCl_2$)
- Halogenoalkane → amine (excess NH_3, sealed tube)
- Halogenoalkane → nitrile (KCN in ethanol/water) → carboxylic acid (acid hydrolysis) or amine ($LiAlH_4$ reduction)
- Alcohol → aldehyde (oxidation, distil) → carboxylic acid (further oxidation, reflux)
- Carboxylic acid + alcohol → ester (acid catalyst)
- Acyl chloride + amine → amide
- Benzene → nitrobenzene (nitration) → phenylamine (reduction with Sn/HCl)

6.7 Analytical Techniques

Mass Spectrometry

Mass spectrometry identifies compounds by their molecular mass and fragmentation pattern. The molecular ion peak (M^+) gives the relative molecular mass. Fragmentation peaks correspond to pieces of the molecule after bonds break. Common fragment losses: 15 (CH_3), 17 (OH), 29 (CHO or C_2H_5), 31 (CH_2OH), 45 (OC_2H_5 or COOH).

The M+1 peak (one mass unit above M^+) is caused by the presence of ^{13}C. The relative height of the M+1 peak indicates the number of carbon atoms in the molecule.



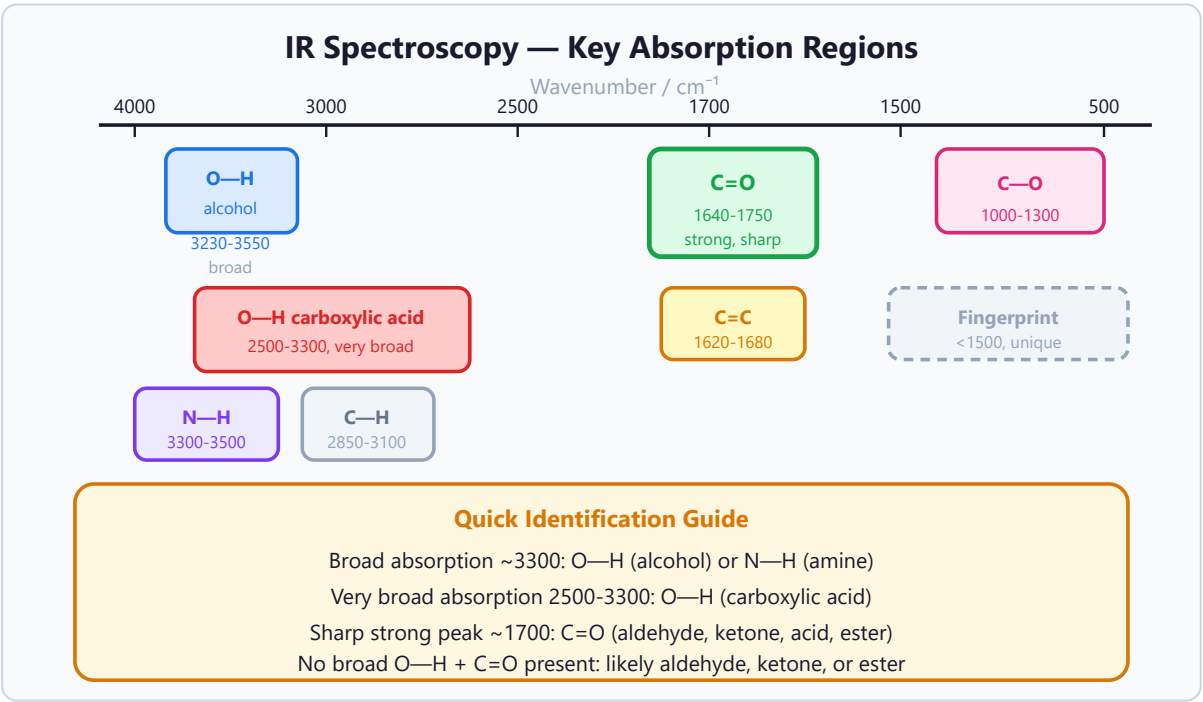
Infrared (IR) Spectroscopy

IR spectroscopy: Identifies functional groups by the absorption of infrared radiation, which causes covalent bonds to vibrate (stretch and bend) at characteristic frequencies. An IR spectrum plots transmittance (%) against wavenumber (cm^{-1}).

Bond	Wavenumber (cm^{-1})	Appearance

O–H (alcohol)	3230–3550	Broad
O–H (carboxylic acid)	2500–3300	Very broad
N–H (amine)	3300–3500	Medium, may show 2 peaks
C–H	2850–3100	Medium to strong
C=O	1640–1750	Strong, sharp
C=C	1620–1680	Medium
C–O	1000–1300	Strong

The fingerprint region (below about 1500 cm^{-1}) is unique to each compound and can be compared against a database for identification. IR spectroscopy can also be used to monitor the progress of a reaction by observing the disappearance or appearance of characteristic absorptions.



NMR Spectroscopy

Nuclear Magnetic Resonance (NMR) spectroscopy: Detects the chemical environments of specific nuclei (usually ^{1}H or ^{13}C) in a magnetic field. Nuclei in different chemical environments absorb at

different frequencies, giving peaks at different chemical shifts (δ, measured in ppm relative to TMS).

Tetramethylsilane (TMS, $Si(CH_3)_4$) is the reference standard (δ = 0 ppm) because it is chemically inert, gives a single sharp peak (all 12 H atoms are equivalent), and its peak is well away from most other signals.

^{13}C NMR

^{13}C NMR shows one peak for each unique carbon environment in the molecule. The number of peaks tells you how many different carbon environments there are. Symmetry can reduce the number of peaks (equivalent carbons give only one peak). Chemical shift values indicate the type of carbon environment:

Carbon environment	Approximate δ / ppm
$R-CH_3$, $R-CH_2-R$ (alkyl)	5–40
C−Cl, C−Br	25–65
C−O (alcohols, esters)	50–80
C=C (aromatic)	110–160
C=O (carbonyl)	170–220

^{1}H NMR

^{1}H NMR is richer in information than ^{13}C NMR because it provides three pieces of data for each signal:

1. **Chemical shift (δ):** Indicates the chemical environment of the hydrogen atoms.
2. **Integration (relative peak area):** Proportional to the number of hydrogen atoms in that environment.
3. **Splitting pattern (multiplicity):** Determined by the number of hydrogen atoms on adjacent carbon atoms (the n+1 rule).

Proton environment	Approximate δ / ppm
$R-CH_3$	0.7–1.2
$R-CH_2-R$	1.2–1.4

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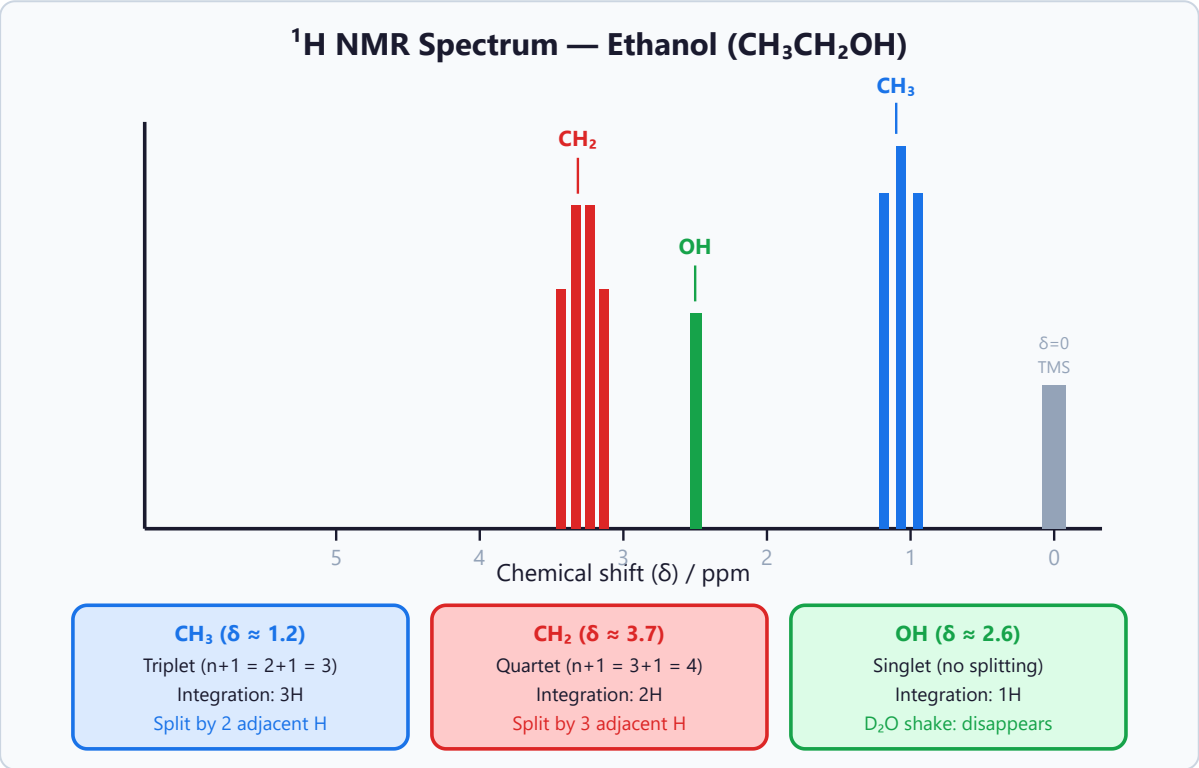
R−CH_2−X (halogen or O)	3.1–4.2
R−O−H (alcohol)	1.0–5.5 (variable)
Ar−H (aromatic)	6.5–8.0
R−CHO (aldehyde)	9.4–10.0
R−COOH (carboxylic acid)	10.0–12.0



Splitting (Spin-Spin Coupling)

The **n+1 rule:** A proton signal is split into (n+1) peaks, where n is the number of equivalent protons on adjacent carbon atoms.

Adjacent equivalent H atoms (n)	Splitting pattern	Name
0	1 peak	Singlet (s)
1	2 peaks	Doublet (d)

2	3 peaks	Triplet (t)
3	4 peaks	Quartet (q)

The relative intensities follow Pascal's triangle: singlet 1, doublet 1:1, triplet 1:2:1, quartet 1:3:3:1. O−H and N−H protons usually appear as singlets (no splitting) because rapid exchange with the solvent averages out coupling. A D_2O shake (adding deuterium oxide) causes O−H and N−H peaks to disappear, confirming their identity.

Exam Tip: When interpreting ^1H NMR spectra: (1) count the number of signals to determine how many different H environments exist; (2) use integration ratios to find the relative number of H atoms; (3) use chemical shifts to identify the environments; (4) use splitting patterns to determine connectivity. Work systematically through all the data before proposing a structure.

6.8 Chromatography

Thin-Layer Chromatography (TLC)

TLC uses a thin layer of silica gel or alumina on a plate (the stationary phase). The sample is spotted near the bottom, and a solvent (mobile phase) rises up the plate by capillary action. Components separate based on their relative affinities for the stationary and mobile phases.

$$R_f = \text{distance moved by substance} / \text{distance moved by solvent front}$$

R_f values are characteristic of a substance under specific conditions and can be compared with standards for identification. TLC is quick and useful for monitoring reaction progress and checking purity.

Gas Chromatography (GC)

GC separates volatile compounds. The sample is vaporised and carried through a long, coiled column by an inert carrier gas (the mobile phase). The column is packed with a high-boiling-point liquid on an inert solid support (the stationary phase). Components separate based on their boiling points and interactions with the stationary phase. A detector records the signal as each component elutes, producing a

chromatogram where peak retention time identifies the component and peak area is proportional to amount.

High-Performance Liquid Chromatography (HPLC)

HPLC is used for non-volatile or thermally unstable compounds that cannot be analysed by GC. A liquid mobile phase is pumped at high pressure through a column packed with fine particles (stationary phase). The technique gives excellent separation and is widely used in pharmaceutical analysis. A UV detector is commonly used. Like GC, retention time identifies components and peak area gives quantitative information.

GC-MS and HPLC-MS combine chromatographic separation with mass spectrometry for both separation and identification in a single analysis, making them extremely powerful analytical tools.

Exam Tip: You may be asked to combine data from mass spectrometry, IR, and NMR to identify an unknown compound. Approach this systematically: (1) use the mass spectrum to find the molecular mass and key fragments; (2) calculate the degree of unsaturation (IHD = (2C + 2 + N − H − X) / 2; IHD of 4 strongly suggests a benzene ring); (3) use IR to identify functional groups; (4) use ^{13}C NMR to count carbon environments; (5) use ^{1}H NMR to determine hydrogen environments, numbers, and connectivity. Propose a structure consistent with all the data.

Worked Example — Structure Determination:

Unknown compound: M^+ = 72. Molecular formula C_4H_8O. IHD = (8 + 2 − 8) / 2 = 1 (one C=O or one ring).
IR: strong sharp absorption at 1720 cm^{-1} (C=O), no broad O–H → ketone or aldehyde, not acid or alcohol.
^{1}H NMR: two peaks only. δ 2.1 (singlet, 6H) and δ no other peak.

6H as a singlet at δ 2.1 suggests two equivalent CH_3 groups next to a C=O with no adjacent H atoms. **Structure: butanone** would give 3 peaks, so this must be **propanone (CH_3COCH_3)** — wait, M_r of propanone = 58 not 72. Reconsider: M_r = 72 with C_4H_8O and IHD = 1. ^{1}H NMR with δ 9.8 (1H, singlet) and δ 0.9 (9H, singlet) would indicate $(CH_3)_3CCHO$ — 2,2-dimethylpropanal. Always cross-check all data against your proposed structure.